

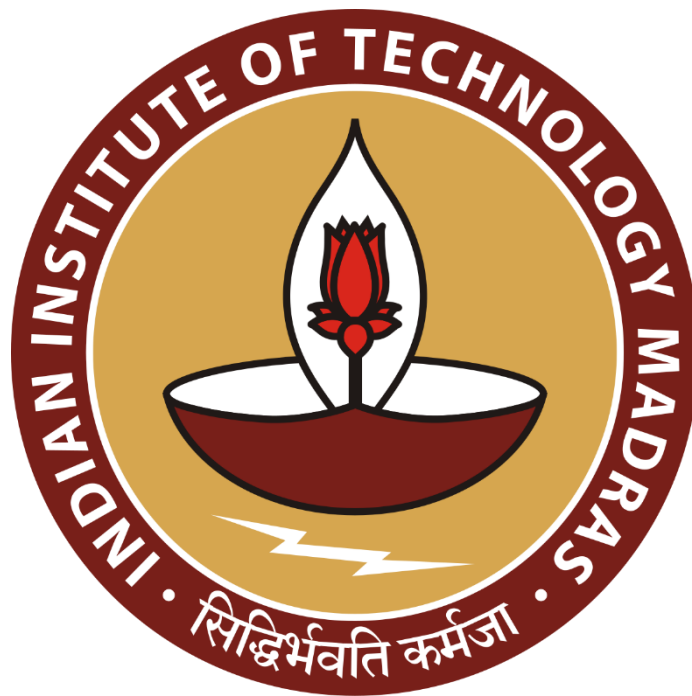
**Service and Operational Optimization of AMRIT Pharmacy at
KGMU, Lucknow**

A Proposal report for the BDM capstone Project

Submitted by

Name: Neeraj Kumar

Roll number: 23f2002096



IITM Online BS Degree Program,
Indian Institute of Technology, Madras, Chennai
Tamil Nadu, India, 600036

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Declaration Statement

I am working on a Project titled “**Service and Operational Optimization of AMRIT Pharmacy at KGMU, Lucknow**”. I extend my appreciation to **Amrit Pharmacy**, for providing the necessary resources that enabled me to conduct my project.

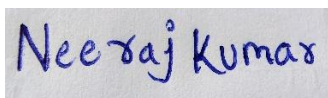
I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analyzed to assure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfillment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.



Signature of Candidate:

Name: Neeraj Kumar

Date: 13/06/2025

1 Executive Summary and Title

Amrit Pharmacy is a state-operated medical dispensary located in Shatabdi Phase 1 at King George's Medical University (KGMU), Lucknow. It aims to provide affordable healthcare by offering prescription medications, over-the-counter (OTC) drugs, and essential health supplies at discounted rates.

Despite its prime location within a major government hospital, the pharmacy faces several challenges like inconsistent stock issues, outdated manual inventory systems, billing delays, and low profit margins on critical items. Moreover, the pharmacy's lack of strong data analysis and marketing efforts, limiting its ability to grow and offer personalized customer service.

This project aims to enhance the pharmacy's operations by employing Excel and Python to analyze purchasing and sales trends, monitor the turnover of high-demand medicines, and identify inefficiencies in procurement and storage processes. By implementing forecasting techniques, inventory control methods, and basic digital engagement strategies (such as SMS-based refill reminders), the goal is to enhance customer satisfaction, minimize stockouts, and increase revenue.

2 Organization Background

Amrit Pharmacy, located in Shatabdi Phase 1 of King George's Medical University (KGMU), started its services on **August 21, 2017**. It was launched through a video conference by the union Health minister at that time, **JP Nadda**.

The pharmacy, established as part of the Amrit (Affordable Medicines and Reliable Implants for Treatment) Pharmacy initiative, is situated within the Shatabdi Phase 1 building of King George's Medical University (KGMU) in Lucknow. It operates on a Business-to-Consumer (B2C) model, offering subsidized medicines to the general public. The pharmacy experiences high demand for life-saving drugs, scheduled drugs, and health supplements.

The pharmacy operates 24/7 with a small staff and serves hundreds of customers daily. Despite its strategic location, the business faces structural and operational inefficiencies that affect customer satisfaction and profit margins.



Amrit Pharmacy Store front image

3 Problem Statement

- 3.1 Stock-Out & Inventory Mismatch:** The pharmacy frequently experiences shortages of high-demand medicines due to inefficient inventory tracking and manual reorder systems.
- 3.2 Billing Inefficiencies & Delays:** Manual or outdated billing systems lead to long queues and customer dissatisfaction, particularly during peak hours.

4 Background of the Problem

Amrit Drug Store serves hundreds of customers daily , but need an advanced system to track medicine turnover in real-time, leading to issues such as expired medicines and overstocking. Staff manually update stock records, resulting in delayed reorders and stock discrepancies.

Furthermore, the pharmacy does not have a customer engagement system to effectively communicate with customers about important updates such as medicine availability, prescription refills, or medicine alternatives. Many customers leave the counter due to long wait times, miscommunication, or lack of transparency about medicine substitutions.

Finally, there is no strategic use of data analytics for identifying fast-moving items, anticipate seasonal demand surges (e.g., during dengue or flu seasons), or categorize products based on profit margins, which is essential for better purchasing decisions.

5 Problem Solving Approach

5.1 Data Collection and Integration

To effectively address the challenges faced by Amrit Pharmacy, the first step is to gather and organize data from various sources. This includes collecting records of daily sales, purchase orders, return logs, expired stock reports, and vendor invoices from the past three months. By organizing this data using tools like Excel, the pharmacy can analyze sales trends, identify fast- and slow-moving items, and understand seasonal demand variations. Additionally, tracking supplier-wise delivery timelines, medicine shelf life, and stock replenishment cycles will assist in making informed procurement decisions. Integrating this data with information on foot traffic during peak hospital hours can help correlate sales performance with external demand factors.

5.2 Analytical Methodology

The next step is to use simple and advanced data analysis on the collected information. Basic analysis with Excel tools like pivot tables and charts will show past patterns, such as which medicines sell the most in certain months, and which types (like antibiotics or emergency drugs) bring in more money. ABC analysis will help sort the stock based on how important and valuable each item is, so the pharmacy can focus on key medicines. Time-series forecasting models will be developed using Python to predict the demand for high-rotation medicines, particularly those sensitive to seasonal illnesses. We'll be using some handy Excel tools, like VLOOKUP and INDEX-MATCH, to match up data from different sheets—such as tracking what's in stock and what's been bought earlier. This helps in figuring out what needs to be ordered again, avoiding piling up too much stuff, and making sure essential items don't run out.

5.3 Tools and Technologies

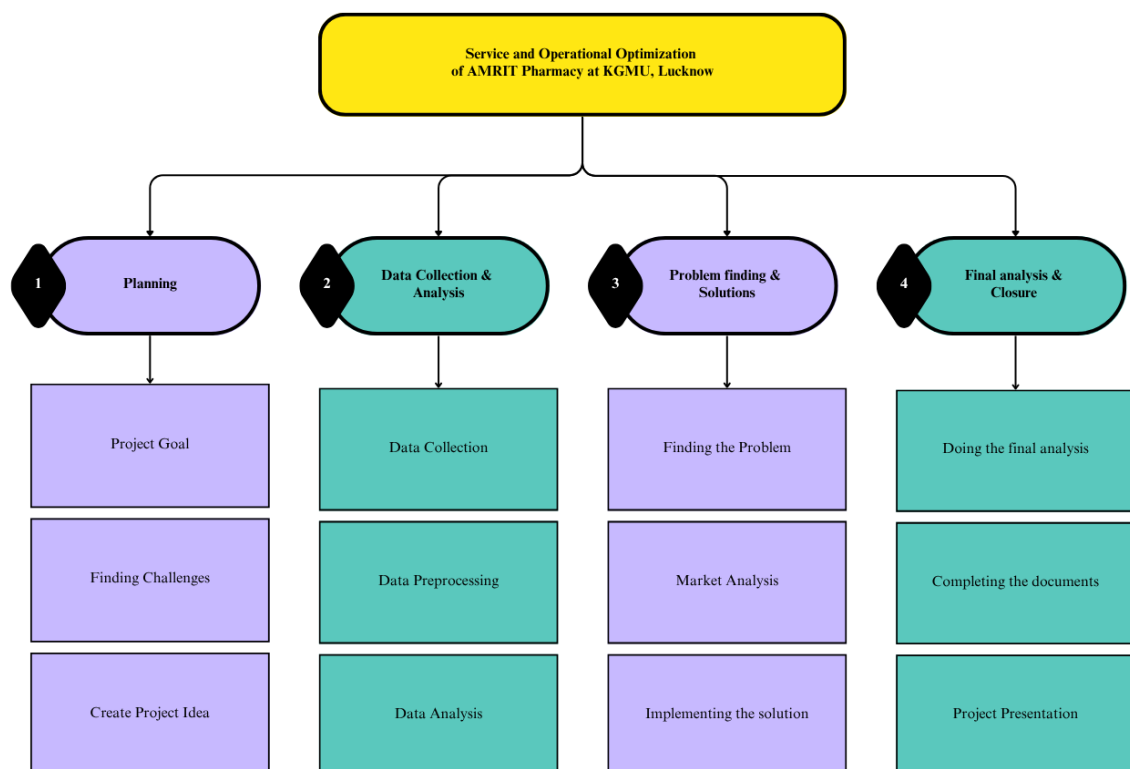
To manage everything smoothly and accurately, we'll rely on a mix of tools. Microsoft Excel will handle most of the stuff—like sorting data, making charts, dashboard and keeping track of inventory with its built-in features. For more detailed work, like predicting which medicines might be needed in the future or running deeper data analysis, we'll turn to Python and its statistical capabilities. These dashboards can display real-time stock updates, notify when supplies are running low, and give a quick view of how well products are selling.

To stay aware of trends outside the pharmacy, we'll also check sources like Google Trends and market reports related to medicines. This can help us compare prices and understand which drugs are in high demand at any given time. For added convenience, we're considering connecting an SMS service (just as a study case) that could send reminders to customers about refilling prescriptions or let them know when essential medicines are back in stock. This kind of communication could really help with customer loyalty.

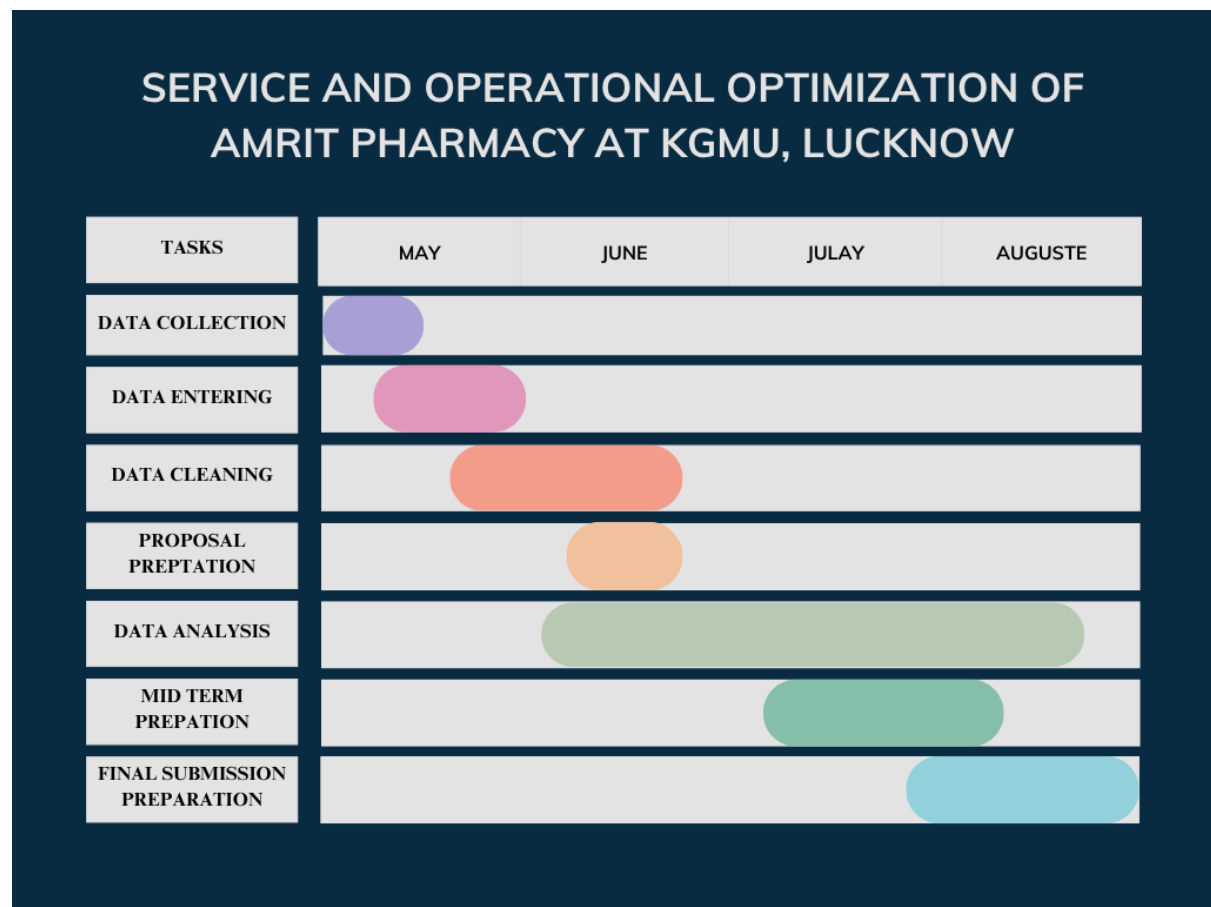
By using these smart, data-driven strategies, Amrit Pharmacy can stay on top of inventory, forecast future needs more accurately, and even try out small-scale digital marketing ideas. Altogether, this approach can boost efficiency, improve customer service, and support the pharmacy's steady growth over time.

6 Expected Timeline

6.1 Work Breakdown Structure:



6.2 Gantt chart



7 Expected Outcome

7.1 Improved Inventory Planning:

Reduction in stockouts and wastage through data-driven reorder points and ABC analysis.

7.2 Reduced Queues & Better Billing

Recommendation for POS upgrade and integration with inventory to improve customer throughput and minimize billing errors.

7.3 Increased Sales through Awareness

Introduction of low-cost SMS alerts and basic promotional boards to increase awareness of available generic/substitute medicines.

7.4 Strategic Procurement

Vendor performance evaluation and discount tracking to increase margins on high-frequency items.